



Challenges in the diagnosis of Parkinson's disease

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Parkinson's disease is the second most common neurodegenerative disease and its prevalence has been projected to double over the next 30 years. An accurate diagnosis of Parkinson's disease remains challenging and the characterisation of the earliest stages of the disease is ongoing. Recent developments over the past 5 years include the validation of clinical diagnostic criteria, the introduction and testing of research criteria for prodromal Parkinson's disease, and the identification of genetic subtypes and a growing number of genetic variants associated with risk of Parkinson's disease. Substantial progress has been made in the development of diagnostic biomarkers, and genetic and imaging tests are already part of routine protocols in clinical practice, while novel tissue and fluid markers are under investigation. Parkinson's disease is evolving from a clinical to a biomarker-supported diagnostic entity, for which earlier identification is possible, different subtypes with diverse prognosis are recognised, and novel disease-modifying treatments are in development.

Introduction

Parkinson's disease is the second most common neurodegenerative disease, with a global prevalence of more than 6 million individuals. This number results from a 2·5-times increase in prevalence over the past 30 years, making Parkinson's disease one of the leading causes of neurological disability.^{1,2} The pathological hallmark of Parkinson's disease consists of neural inclusions in the form of Lewy bodies and Lewy neurites, with cell loss in the substantia nigra and other brain areas. Given that aggregated and misfolded α -synuclein species are the major constituents of Lewy bodies, Parkinson's disease is classified as a synucleinopathy. Braak and colleagues³ have proposed a pattern of spread of Lewy pathology, starting in the caudal brainstem and progressing rostrally through the upper brainstem, limbic regions, and finally the neocortex, but such spread probably does not occur in all cases. Prion-like, cell-to-cell transmission, and permissive templating of synuclein are potential mechanisms of disease progression.⁴

Age is the most important risk factor for developing Parkinson's disease, and men are more susceptible than women, with a prevalence ratio of approximately 3:2. A strong genetic component to disease risk has been identified, with more than 90 associated loci.⁵ Additionally, several potential modifiable environmental (eg, pesticides, water pollutants) and other factors (eg, smoking, coffee, exercise, or head trauma)¹ have been found to play a role in the pathogenesis of Parkinson's disease in different populations. Although these advances in our understanding of pathogenesis and epidemiology have been substantial,^{6,7} the cause of Parkinson's disease remains enigmatic, and no cure or preventive therapy has yet been found.

Clinical diagnostic criteria, designed to enhance the diagnostic accuracy of Parkinson's disease, have been validated over the past 5 years. However, diagnosis remains a challenge because clinical features can overlap with those of other neurodegenerative conditions, and tests or biomarkers do not allow for a definitive diagnosis from the earliest stages. As a result, clinical diagnostic accuracy remains suboptimal, even when the disease is clinically

fully manifested.⁸ Identification of prodromal disease is needed, given that future disease-modifying therapies will have their greatest chance for success at this stage.^{9,10} Finally, there is also a need to better define Parkinson's disease subtypes,^{11–13} which not only have different clinical presentation and prognosis, but also differ in underlying disease mechanisms, calling for personalised treatment approaches. The most obvious example is monogenic Parkinson's disease, for which subtype-specific therapies are already being tested in clinical trials.^{14,15}

This Review, directed towards general neurologists and movement disorder specialists involved in the diagnosis and care of patients with Parkinson's disease, and neuroscientists, describes the motor and non-motor features of Parkinson's disease, and delineates the issues involved in identifying disease subtypes and the increasing role of genetics in their diagnosis. We will also discuss the challenges encountered when diagnosing the manifest and so-called premotor stages of the disease and critically review the imaging, fluid, and tissue biomarkers that best support the diagnosis of Parkinson's disease.

More than a movement disorder

The clinical hallmark of Parkinson's disease is a motor syndrome characterised by bradykinesia, rest tremor, and rigidity, in addition to changes in posture and gait. The motor disturbances cause progressive disability, with impairment in activities of daily living and reduced quality of life. Although the classic motor symptoms occur early and are the pillars of diagnostic criteria, the development of postural instability and increasing gait difficulties, as well as dysphagia and dysarthria, drive the progression of motor disability.¹⁶

Parkinson's disease is considered a movement disorder, but is associated with a variety of non-motor symptoms in virtually all patients, including hyposmia, constipation, urinary dysfunction, orthostatic hypotension, memory loss, depression, pain, and sleep disturbances (panel 1). The classic motor signs of Parkinson's disease are linked to nigral degeneration and striatal dopamine depletion, whereas non-motor symptoms are probably associated with neurodegeneration of other structures, including the

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For more on **The University of Pennsylvania Smell Identification Test** see <https://sensonics.com/product/smell-identification-test/>

Panel 1: Motor and non-motor symptoms of Parkinson's disease

Motor features occurring at early stages

These features are considered the so-called classic or cardinal motor features of the disease.

Bradykinesia

General slowness and paucity of spontaneous movement; decreased arm swing, reduced facial expression, reduced gesticulation, micrographia, difficulties turning in bed, and hypophonia. Progressive reduction in speed and amplitude of voluntary repetitive movement (finger taps, hand grips, pronation-supination movements, toe taps, and heel stamps).

Rigidity

Increased muscular tone by a resistance of passive movements of equal degree in opposing muscle groups (so-called lead-pipe type). When the increased muscle tone is interrupted by a co-existing tremor, a ratchet-like quality of resistance, named cog-wheel rigidity, is felt.

Tremor

Rest tremor (4–6 Hz) is common in limbs (so-called pill-rolling in hands), lips, chin, or jaw, and rarely in the head. Amplitude diminishes or is abolished during goal-directed voluntary movements; hand rest tremors are examined with hands in a relaxed position and arms supported—for example, hands folded into the lap while sitting, and forearms in pronation (not supination). Low amplitude hand action tremor is also common at presentation.

Gait alterations

Decreased arm swing, dragging one leg, and slightly bent posture while walking.

Motor features present at later stages

These motor features generally occur in addition to earlier ones and respond poorly to dopaminergic treatment.

Posture alterations

Trunk bent forward when standing. Lateral (so-called Pisa syndrome) or anterior (camptocormia) deviation of trunk, or head flexion (so-called dropped head). Arms abducted, flexed at elbow. Flexed wrist and metacarpophalangeal joints, and extended hand fingers and thumb.

Freezing of gait

Sudden and brief episode of inability to produce effective forward stepping at initiation of gait (so-called start hesitation), during gait (motor block), or when turning or approaching narrow spaces. Festination occurs when patients are compelled to accelerate the gait forward.

Balance alterations

Unsteadiness when standing and walking. Altered postural reflexes (tested with the pull test), and falls.

Other

Dysarthria and dysphagia.

Non-motor features present at early stages

These are not uncommon at the time of diagnosis and can precede the onset of motor features.

Hyposmia

Smell loss is reported by up to 70% of patients, and when formally tested, presents in almost 90% of patients. Frequent smell tests used are The University of Pennsylvania Smell Identification Test and the Sniffin Sticks test.

Sleep disorders

Rapid eye movement sleep behavior disorder is a parasomnia characterised clinically by vivid, generally unpleasant dreams (eg, being attacked or robbed) and vigorous behaviours in which the patients seem to be enacting their dreams (eg, punching, shouting, laughing). Additionally, sleep signs and symptoms can include insomnia, periodic limb movements, restless leg syndrome, akathisia, and excessive daytime sleepiness.

Neuropsychiatric features

Prominent apathy; anxiety (generalised anxiety, panic attacks, and social phobias); and depression (usually mild and associated to anhedonia and apathy)

Autonomic dysfunction

Constipation, delayed gastric emptying, urinary urgency or incontinence, erectile dysfunction, orthostatic hypotension, and heat intolerance.

Mild cognitive impairment

Mild cognitive decline in executive and attention domains.

Pain and somatosensory disturbances

Pain, paraesthesias, and burning sensations.

Non motor features present at later stages

Early non motor features frequently persist and worsen at this stage.

Dementia

About 30% of patients develop dementia, affecting visual spatial recognition and construction, and semantic and episodic memory. Prevalence increases with disease duration. Frequently associated with hallucinations and psychosis.

peripheral autonomic nervous system.^{17,18} Non-motor symptoms are frequent at early stages and, although intense and disturbing for some patients, observational studies indicate that they are mild in most cases,^{19,20} increasing in severity with disease duration.¹⁹ As the disease progresses, non-motor symptoms cause an increased burden, reduce quality of life, and are a driver of the overall

cost of care.²¹ In particular, cognitive decline and hallucinations are common causes of admission to hospital and institutionalisation in advanced stages of the disease.²²

The prodromal stage

Several non-motor symptoms associated with Parkinson's disease, such as smell loss or constipation, are commonly

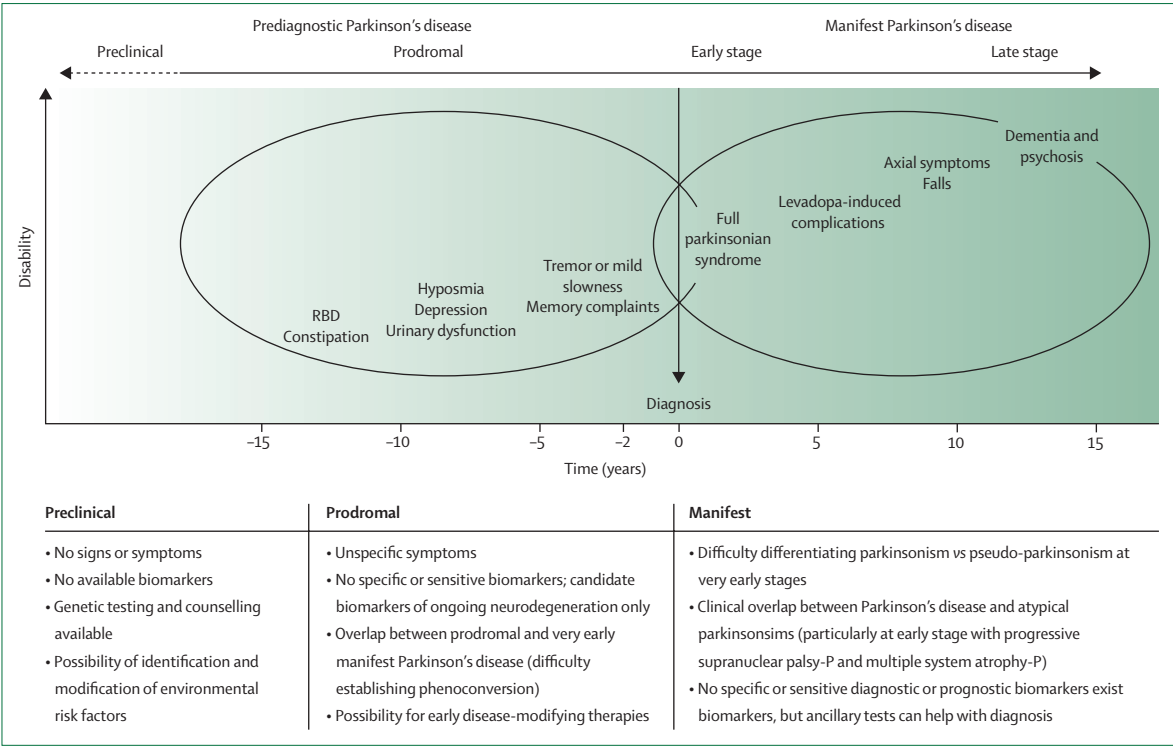


Figure: The natural history of Parkinson's disease and diagnostic challenges by disease stage
RBD=REM sleep behavior disorder. Progressive supranuclear palsy-P=progressive supranuclear palsy with predominant parkinsonism. Multiple system atrophy-P=multiple system atrophy with predominant parkinsonism. The time of diagnosis is represented in the axis as time "0". The timepoints on the left side of diagnosis represent the number of years before diagnosis, and the timepoints on the right represent the years after diagnosis. These periods of time are orientative. The dotted arrow indicates that the duration of the preclinical phase is unknown, unlike the prodromal phase, which can extend between 10 and 15 years.

reported by patients before the onset of classic motor symptoms—sometimes preceding the occurrence of motor features by years or even decades.²³ The period when these symptoms arise has been conceptualised as the prodromal phase of Parkinson's disease, corresponding to a stage of disease when neurodegenerative changes involve extranigral sites, such as the lower brainstem, the olfactory bulb and tracts, and the peripheral autonomic nervous system (Braak stages 1–3).³ Similar to Alzheimer's disease, an even earlier period when future patients are still free of any symptoms, but disease-specific pathology is assumed to be present and there is biomarker evidence of disease, has also been postulated for Parkinson's disease (termed preclinical Parkinson's disease; figure).²⁴

The evidence that some non-motor symptoms are markers of a prodromal phase of Parkinson's disease is based on retrospective assessments as well as prospective epidemiological and observational studies,^{23,25} and is most compelling for constipation, smell loss, and rapid eye movement-sleep behaviour disorder. Additionally, urinary urgency, sexual dysfunction, hypotension, anxiety, depression, colour vision impairments, and dysexecutive syndrome have also been described to antedate the onset of motor symptoms in Parkinson's disease (figure).^{19,23,25} Important issues about the prodromal phase of Parkinson's

disease remain to be clarified, such as the sequence in which prodromal symptoms develop and the speed of disease progression.²⁶ Prodromal features can also vary depending on etiology (eg, idiopathic vs monogenic Parkinson's disease).²⁷ In addition to non-motor symptoms, subtle motor signs such as decreased facial mobility, voice changes, loss of finger dexterity, a mildly stooped posture, or decreased arm swing when walking might also antedate the evolution of definitive motor symptoms. However, such mild parkinsonian signs might be difficult to distinguish from unspecific mobility changes associated with normal ageing.²⁸

Disease subtypes

Parkinson's disease is strikingly heterogeneous regarding the age of onset, clinical presentation, rate of progression, and treatment response. Several clinical subtypes have been proposed. Additionally, the discovery of genetically defined forms of the disease, which can differ from sporadic forms in a number of clinical variables, have challenged the unitarian view of Parkinson's disease and introduced a biological definition of subentities within the Parkinson's disease spectrum.

Approaches towards subtyping Parkinson's disease have either used empirical assessments of individual clinical

features or the more objective and hypothesis-free methodology of hierarchical cluster analysis and other forms of machine learning.^{12,13,29} Clinical features that have been used for subtyping with either approach included age at onset (early-onset vs late-onset), prevailing motor phenotype (tremor-dominant vs non-tremor cases), motor complications in response to levodopa treatment, non-motor features (particularly autonomic dysfunction, cognitive dysfunction, and REM sleep behavior disorder [RBD]), as well as their rate of progression.

Empirically defined subtypes include young-onset Parkinson's disease or early-onset Parkinson's disease, usually defined by age at onset cutoffs younger than 40 or 50 years and characterised by slower progression, preserved cognition, and increased risk to develop motor complications in response to levodopa.^{30,31} Benign-tremulous Parkinson's disease or tremor-dominant Parkinson's disease are two terms that have been used to describe the clinical predominance of rest tremor over other motor symptoms³² and this clinical subtype has been associated with slower progression and less cognitive decline than other clinical presentations.^{31–33} Clinical presentations with prominent postural instability and gait disorder have been classified as a postural instability and gait disturbances subtype, characterised by a rapid decline of motor function as well as cognition.³³ Problems with empirically defined subtypes include the fact that patients initially presenting with tremulous or non-tremulous Parkinson's disease motor signs can change categories with longer follow-up.^{33,34}

Recent cluster analyses have included non-motor features,³⁵ and in one of these studies, mild cognitive impairment, RBD, and orthostatic hypotension at baseline identified the most rapidly progressive subtype,³⁶ which was termed diffuse malignant because of the most severe expression of both motor and non-motor features. The slowest progression was seen in patients presenting with predominant motor features of mild severity (mild motor-predominant), with a third subtype being termed intermediate (between the two).³⁶

The ultimate proof for the validity of clinically defined disease subtypes should come from objective biological measures or biomarkers, showing that such sub-entities reflect differences in underlying disease mechanisms or pathology.²⁹ However, a brain bank study in which 111 patients had been retrospectively classified into mild motor-predominant, intermediate, and diffuse malignant subtypes found no group differences in Lewy pathology and Alzheimer's-related pathology.³⁷ Only genetic subtyping of Parkinson's disease has established biological underpinnings.

Lessons from genetics

The advent of the genomics era has led to rapid advances in our understanding of the genetic causes and risk variants of Parkinson's disease. These discoveries have been driven by improvements in sequencing and

genotyping technology, and their successful application to ever-larger cohorts. International efforts have revealed that the genetic architecture of Parkinson's disease is highly complex, with both common and rare risk variants contributing to pathogenesis.⁵ Mutations in at least 20 genes are recognised as causes of familial parkinsonism, each providing a snapshot into the molecular basis of the neurodegenerative process. Notably, more than 90 genetic risk loci for the more common sporadic form of the disease have been identified.³⁸ Although unravelling the precise biology disrupted by these variants is challenging, the disease-associated genes begin to coalesce into common pathways, including dysregulation of mitochondrial homeostasis and impaired processes related to cell death machinery, inflammatory signalling, intracellular trafficking, and endosomal-lysosomal dysfunction.³⁹

Genetic testing for Mendelian forms of Parkinson's disease is increasingly done in clinical practice, and should be considered in patients with early onset of disease (defined as onset before age 40 years), patients with a family history, and individuals from high-risk populations with a high prevalence of specific monogenic forms of disease (eg, Ashkenazi Jewish, North African Berber Arabs).⁴⁰ Knowledge of the underlying gene defect within a family enables more effective counselling and allows for predictive testing within asymptomatic family members. Increasingly, clinical trials are targeting specific genetic forms of neurodegeneration, and the identification of the causative gene can provide opportunities for the patient to participate in such studies.

Genetic information is refining our fundamental understanding of the clinical entity known as Parkinson's disease. An early lesson learned from studying monogenic patients is that Parkinson's disease is phenotypically diverse, and there is more substantial overlap with atypical parkinsonism than previously known. For example, patients harboring a disease-causing mutation in the *LRRK2* gene can manifest with protean clinical presentations that include typical levodopa responsive Parkinson's disease in the majority of cases, progressive supranuclear palsy, and occasionally amyotrophy.^{15,41,42} Similarly, patients with mutations in the genes *GBA*, *SNCA*, or *VPS13C* can present with typical Parkinson's disease, but more commonly develop progressive cognitive impairment consistent with Lewy body dementia.^{43–45} Although these observations only relate to 5–40% of Parkinson's disease cases (depending on ethnic background), these findings provide crucial insights into the central pathways associated with parkinsonism, and emphasise potential targets for disease-modifying interventions. Although monogenic Parkinson's disease cases are increasingly defined on a molecular basis (eg, PARK-*LRRK2*, PARK-*SNCA*), only PARK-*LRRK2* and PARK-*Parkin* are relatively common in clinical practice, as is Parkinson's disease associated with high-risk variants in *GBA*⁴¹ (table 1).

Mutations		Inheritance	Clinical phenotype relative to classic Parkinson's disease			Frequency
			Age at onset	Clinical features	Progression	
Classic parkinsonism						
PARK-SNCA (PARK 1, PARK 4)	Missense (PARK 1); duplication or triplication (PARK 4)	Autosomal dominant	Younger	Similar, prominent non-motor symptoms, and early dementia	Faster	Rare
PARK-LRRK2 (PARK 8)	Missense	Autosomal dominant	Similar	Similar (lower prevalence of rapid eye movement sleep behavior disorder)	Similar	Common in Ashkenazi Jews, Basque population (Spain), and North African Berbers
PARK-VPS35 (PARK 17)	Missense	Autosomal dominant	Similar	Similar	Similar	Rare
PARK-CHCHD2 (PARK 22)	Missense, splice site, nonsense	Autosomal dominant	Similar	Similar	Similar	Rare and controversial subtype, most often described in Asian patients
Early-onset parkinsonism						
PARK-Parkin (PARK 2)	Missense, loss-of- function, exonic duplication, deletion	Autosomal recessive	Younger	Common early leg involvement and dystonia at onset, and frequent dyskinesia	Slower	Common (accounts for up to 20% of cases with onset before age 50 years; dementia uncommon)
PARK-PINK1 (PARK 6)	Missense, loss-of- function, exonic deletion, duplication	Autosomal recessive	Younger	Similar to PARK-Parkin, with common psychiatric features	Slower	Second most common recessive Parkinson's disease subtype after PARK-Parkin; dementia uncommon
PARK-DJ1 (PARK 7)	Missense, loss-of- function, small duplications or deletions	Autosomal recessive	Younger	Similar to PARK-Parkin, with psychiatric symptoms	Slower	Rare; dementia uncommon
High-risk gene						
GBA	Missense, splice site, loss-of-function, small insertions or deletions	Risk gene	Younger	Similar to sporadic Parkinson's disease, but greater dementia risk	Faster	Very common (5–25% of patients), particularly in Ashkenazi Jews

Table 1: Monogenic subtypes of Parkinson's disease

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Diagnostic challenges and pitfalls

Making a diagnosis of idiopathic Parkinson's disease can be a straightforward clinical exercise in cases with a classic history, typical asymmetric motor signs, no atypical features, and exclusion of alternative causes.

However, in routine clinical practice diagnostic misclassification is common, with error rates ranging from 15% to 24% in different series.^{8,46,47} A recent meta-analysis found a pooled diagnostic accuracy for the clinical diagnosis of Parkinson's disease of only 80·6% across 11 clinico-pathological studies.⁴⁸ Even with the use of stringent clinical diagnostic criteria, 10% of people diagnosed with Parkinson's disease by neurologists had alternative pathologies. Common misdiagnoses in clinical practice include non-Parkinson's disease tremor disorders, such as essential tremor, as well as different types of secondary parkinsonism (table 2).

The greatest challenge, even for movement disorder specialists, is early diagnostic differentiation of Parkinson's disease from atypical parkinsonian disorders. Atypical parkinsonism is an umbrella term for a variety of neurodegenerative disorders in which a parkinsonian syndrome is a prominent clinical feature, but the full clinical spectrum, underlying pathology, progression, and prognosis fundamentally differ from those of Parkinson's disease. The atypical parkinsonism syndromes include multiple system atrophy, which is pathologically defined

by glial cytoplasmic inclusions of misfolded α -synuclein in oligodendrocytes, as well as the tauopathies progressive supranuclear palsy and corticobasal degeneration, defined by neuronal deposition of four-repeat phosphorylated tau aggregates.^{49,50} At early stages of the disease, all three conditions can be very difficult to distinguish from Parkinson's disease, as well as from each other. Clinico-pathological studies have revealed error rates in clinical diagnosis in 7–35% of cases.^{51–53}

Clinical pointers that can inform the differential diagnosis between Parkinson's disease and these main types of atypical degenerative parkinsonism are summarised in panel 2. These differentiating features can evolve over time and, particularly the parkinsonian variants of multiple system atrophy and progressive supranuclear palsy, can be notoriously difficult to distinguish from Parkinson's disease in early disease stages—including asymmetry (which is particularly striking in corticobasal degeneration) and responsiveness to levodopa.

Clinical diagnostic criteria

To enhance diagnostic accuracy of Parkinson's disease, the International Parkinson and Movement Disorder Society has proposed a set of criteria that represent a revised version of the Queen's Square Brain Bank Criteria that have been the most commonly used over the past decades.^{54,55} These criteria rest on the expert neurological

	Mechanism	Differential clinical features vs Parkinson's disease	Diagnosis	Therapy
Drug-induced*	Interference with dopamine signalling	Often symmetric, with perioral tremor, and co-existent tardive syndromes	Consistent history of exposure; normal dopamine transporter-SPECT	Discontinue the causative drug; temporary use of anti-Parkinson's disease drugs
Vascular	Disruption of striato-pallido-thalamo-cortical motor network	Acute or subacute onset (not obligatory); often presenting with gait disorder (lower body parkinsonism)	Strategic infarcts and subcortical microvascular lesions on MRI, and normal dopamine transporter-SPECT (not obligatory)	Trial of levodopa; physiotherapy; occupational therapy
Toxic (copper, manganese)	Basal ganglia lesions (putamen, pallidum)	Symmetric parkinsonism; co-existent dystonia; severe dysarthria, so-called cock-gait (manganese)	History of exposure; MRI findings	Trial of levodopa; physiotherapy; speech therapy; occupational therapy
Infectious	Basal ganglia abscesses or granuloma (toxoplasmosis, cryptococcosis, tuberculosis); encephalitic (HIV, Creutzfeldt-Jakob disease, progressive multifocal leuko-encephalopathy) or postencephalitic basal ganglia involvement	Additional movement disorders and other neurological signs are common	Medical history; systemic signs; MRI findings; CSF analysis; specific serologies	Treatment of underlying conditions; trial of levodopa
Autoimmune	Anti-neuronal antibodies affecting basal ganglia motor circuits (eg, D2R, DPPX, NMDAR, IgLON5, and Ma2/Ta antibodies)	Additional movement disorders and other neurological signs common	Antibody detection; search for associated neoplasms	Immunotherapy (intravenous immunoglobulin, plasmapheresis, immunosuppressants); treatment of associated tumour
Neoplastic	Invasion or indirect compressive effects (frontal meningioma) of basal ganglia circuitry	Additional focal neurological signs	MRI	Treatment of underlying conditions; trial of levodopa
Metabolic	Basal ganglia involvement (eg, Wilson's disease, non-ketotic hyperglycaemia, extrapontine myelinolysis, calcium dyshomeostasis, hypermagnesaemia in liver disease, iron deposition in NBIA)	Additional movement disorders and other neurological, psychiatric, and systemic signs are common	Specific laboratory and imaging studies	Treatment of underlying conditions; trial of levodopa
Normal pressure hydrocephalus	Compromised prefrontal motor connectivity	Small step and broad-based gait disorder with freezing, no rest tremor, or upper limb involvement (lower body parkinsonism)	Neuroimaging (brain CT or MRI)	Cerebrospinal fluid drainage (repeated lumbar puncture, ventricular shunting)
Functional	Multifactorial, includes psychiatric comorbidity and impaired self-agency	Abrupt onset; spontaneous fluctuation; effortful demonstrative slowness; tremor with frequency variation and entrainment; no response to levodopa	History of psychiatric comorbidity; incongruent clinical presentation; remission with behavioural or psychotherapy	Counselling; cognitive behavioural therapy

NBIA=neurodegeneration with brain iron accumulation. SPECT=single-photon emission computed tomography. *Most common causative drugs: dopamine receptor blockers, including first generation (phenothiazines and butyrophenones) and second generation (eg, olanzapine, risperidone, sulpiride, aripiprazole) antipsychotics, as well as antiemetics (metoclopramide, prochlorperazine, and trifluoromazine); dopamine-depleting drugs (tetrabenazine or reserpine); calcium channel antagonists (flunarizine, cinnarizine, and verapamil); antiepileptics (valproate, carbamazepine, or lamotrigine); antidepressants (selective serotonin reuptake inhibitors, combined noradrenergic-serotonergic reuptake inhibitors, and antimuscarinics).

Table 2: Secondary parkinsonisms

examination showing parkinsonian syndrome defined by the presence of bradykinesia and at least one additional cardinal motor feature (rigidity or classic asymmetric 5-Hz resting tremor), plus the application of supportive and exclusionary features. By contrast to the Queen's Square Brain Bank criteria, the International Parkinson and

Movement Disorder Society criteria list a number of non-exclusionary clinical features that are unusual in Parkinson's disease and should raise suspicion of potential alternative diagnoses (so-called red flags). On the basis of the presence of supportive and absence of exclusionary features, as well as the presence or absence of red flags, the

Movement Disorder Society criteria operationalise two levels of diagnostic certainty for Parkinson's disease, namely clinically established and clinically probable. The first category establishes a set of criteria aimed to maximise specificity at the possible expense of sensitivity, while criteria for the second level aim for enhanced sensitivity (appendix).

A validation study of the Movement Disorder Society criteria has shown excellent sensitivity (96%) and specificity (95%) for a diagnosis of clinically probable Parkinson's disease. The specificity of a diagnosis of clinically established Parkinson's disease was even higher (98.5%), but—as anticipated—this was at the expense of reduced sensitivity (59.3%). For patients with a disease duration of less than 5 years, the specificity of a clinically probable Parkinson's disease diagnosis was 87%.⁵⁶ The International Parkinson and Movement Disorder Society criteria incorporate two ancillary tests among the four supportive diagnostic criteria, but future diagnostic algorithms will need to incorporate additional tests and biomarkers to further enhance diagnostic accuracy and sensitivity for early and prodromal disease stages.

Progress towards early diagnosis

There is consensus that the process leading to clinically defined Parkinson's disease starts much earlier than can be captured by current diagnostic criteria (figure). To date, no biomarkers have been identified that would enable a confident diagnosis of any of these conceptual phases of pre-diagnostic Parkinson's disease with high sensitivity and specificity. This is particularly problematic when it comes to the prognosis of individuals who present with one or more features associated with an increased risk of developing clinically defined Parkinson's disease, such as a positive family history or asymptomatic carrier status for disease-associated mutations plus non-motor features of prodromal Parkinson's disease like hyposmia or RBD.

In a research setting, identifying individuals at risk for Parkinson's disease is important to understand disease progression and to recruit participants for clinical trials of disease-modifying therapies.²⁶ Several cross-sectional and prospective case-control studies have attempted to define the predictive value of prodromal clinical, non-genetic, and genetic risk factors, and neuroimaging tests, to determine the probability of conversion to clinically manifest Parkinson's disease.^{23,25,28} RBD stands out among the clinical markers of Parkinson's disease risk in that more than 90% of individuals with isolated RBD will eventually develop neurodegenerative parkinsonism—most commonly Parkinson's disease or Parkinson's disease dementia.^{57,58} The latent period from RBD onset to the development of Parkinson's disease is variable and usually more than 10 years,⁵⁷ although the presence of olfactory dysfunction, abnormalities on dopamine-transporter SPECT, or transcranial sonography have been shown to identify those patients close to clinical conversion.^{58,59}

Panel 2: Clinical diagnostic pointers for atypical parkinsonism

Multiple system atrophy

- Poor response to levodopa (initial responsiveness to levodopa in about 30% of cases)
- Severe and early autonomic failure (orthostatic hypotension, erectile dysfunction, post-void residual volume >100 mL) in the first 5 years of disease
- Nocturnal stridor
- Early marked dysarthria
- Rapid disease progression
- Early postural instability
- Babinski sign or other pyramidal signs
- Cerebellar signs
- Jerky postural hand tremor (minipolymyoclonus)
- Disproportionate antecollis (so-called dropped head)
- Orofacial dystonia induced by levodopa

Progressive supranuclear palsy

- Akinetic rigid parkinsonism with poor levodopa response
- Slowing of vertical saccades
- Supranuclear downgaze palsy (often not present in the first year of onset)
- Square wave jerks
- Levator inhibition
- Blepharospasm
- Pseudobulbar crying
- Early dysarthria and dysphagia
- Early postural instability and falls
- Early progressive gait freezing
- Early marked frontal dementia

Corticobasal degeneration

- Levodopa-resistant unilateral akinetic parkinsonism
- Cortical sensory loss (ie, agrophesthesia, astereognosia with intact primary sensory modalities)
- Limb apraxia, alien limb phenomenon
- Focal arm myoclonus or dystonia
- Early cognitive impairment with frontal signs and language problems (ie, progressive, non-fluent aphasia)
- Early postural instability and falls

See Online for appendix

The International Parkinson and Movement Disorder Society has proposed criteria for a research diagnosis of prodromal Parkinson's disease. These attempts provide an evidence-based framework to statistically estimate the likelihood for future Parkinson's disease at an individual level on the basis of a large set of well-characterised markers of risk.^{60,61} Prospective cohort-studies have provided evidence for the validity of these criteria in population-based cohorts.^{62,63} A similar algorithm has been developed in the format of an online tool to assess Parkinson's disease risk and has been tested and validated in the PREDICT-Parkinson's disease study, a prospective community-based population study involving more than 1000 participants.^{64,65}

Panel 3: Useful diagnostic tests in patients presenting with parkinsonism**UPSIT and Sniffin Sticks for olfactory function**

- Normosmia: questions a Parkinson's disease diagnosis
- Hyposmia: consistent with Parkinson's disease

Imaging*Structural MRI*

- Normal or signs of unrelated co-morbidity: consistent with Parkinson's disease
- Structural basal ganglia pathology (eg, infarcts, haematoma, abscess, calcification, iron deposition), frontal meningioma, normal pressure hydrocephalus: secondary parkinsonism
- Putamenal atrophy and hypointensity, putamenal rim sign, pontocerebellar atrophy, middle cerebellar peduncle atrophy, hot cross bun sign: suggestive of multiple system atrophy
- Midbrain atrophy (hummingbird sign), dilated third ventricle, superior cerebellar peduncle atrophy: suggestive of progressive supranuclear palsy
- Asymmetric parietal cortical atrophy: suggestive of corticobasal degeneration

Diffusion-weighted-MRI

- Normal: consistent with Parkinson's disease
- Increased putamenal diffusivity: suggestive of multiple system atrophy (can also be seen in progressive supranuclear palsy)
- Increased diffusivity in middle cerebellar peduncle: suggestive of progressive supranuclear palsy

Dopamine transporter-SPECT

- Abnormal (asymmetric reduction of striatal tracer binding): consistent with Parkinson's disease or other degenerative parkinsonism

- Normal: excludes Parkinson's disease or other degenerative parkinsonism

Meta-iodobenzylguanidine-SPECT

- Reduced cardiac meta-iodobenzylguanidine uptake: consistent with Parkinson's disease (inconclusive in early disease)
- Normal: suggestive of non-Parkinson's disease parkinsonism (inconclusive in early disease) or secondary parkinsonism

Fluorodeoxyglucose PET

- Putamenal hypermetabolism (plus occipital and parietal hypometabolism): consistent with Parkinson's disease
- Putamenal and cerebellar hypometabolism: consistent with multiple system atrophy
- Frontal, caudate, and brainstem hypometabolism: consistent with progressive supranuclear palsy
- Asymmetric striatal and parietal hypometabolism: consistent with corticobasal degeneration

Transcranial ultrasound

- Midbrain hyperechogenicity and basal ganglia normoechogenicity: consistent with Parkinson's disease
- Midbrain normoechogenicity and basal ganglia hyperechogenicity: might be suggestive of non-Parkinson's disease parkinsonism

Genetic testing

- Pathogenic mutation in known Parkinson's disease gene: confirms genetic Parkinson's disease
- Absence of a pathogenic mutation in known Parkinson's disease genes: does not rule out Parkinson's disease

Although current research algorithms might provide an opportunity for earlier detection of Parkinson's disease than is currently possible in clinical practice, their sensitivities and predictive values are still suboptimal, and sensitive and reliable Parkinson's disease biomarkers are urgently required.

Diagnostic testing—from clinical routine to future biomarkers

Nowadays, clinicians must rely on the judicious use of a limited number of diagnostic tests to solidify a clinical diagnosis of Parkinson's disease. Their use follows principles of cost-effectiveness, and diagnostic yield is context dependent. We summarise broadly available ancillary tests that have been established to support Parkinson's disease or an alternative diagnosis (panel 3).

Olfactory function testing using The University of Pennsylvania Smell Identification Test or Sniffin Stick tests has been extensively studied in Parkinson's disease and other parkinsonian syndromes. Hyposmia or anosmia have been consistent findings in about 90% of patients with Parkinson's disease, while normosmia is

the rule in the early stages of atypical degenerative or secondary parkinsonisms.⁶⁶ The International Parkinson and Movement Disorder Society criteria for Parkinson's disease lists hyposmia as one of four supportive criteria of a Parkinson's disease diagnosis, and although in their validation study⁶⁶ the olfactory testing only achieved 63·4% specificity, this feature has shown high diagnostic accuracy for distinguishing Parkinson's disease from multiple system atrophy and progressive supranuclear palsy in other studies.⁶⁷ Given the low cost and easy applicability, olfactory testing should be part of the initial clinical testing of people with suspected Parkinson's disease.

Imaging markers

Structural MRI is usually unremarkable in patients with Parkinson's disease. Nonetheless, it should also be part of the routine diagnostic process to distinguish Parkinson's disease from secondary or atypical parkinsonian syndromes (panel 3) because several MRI features are highly specific for atypical parkinsonisms, although sensitivity is low (at about 50%).

Novel MRI techniques, including neuromelanin imaging (NMI), quantitative susceptibility mapping (QSM), or visual assessment of dorsal nigral hyperintensity, have the potential to assess nigral pathology and have been a major focus of research efforts. NMI exploits the paramagnetic properties of neuromelanin, while QSM enables quantification of iron deposition in the substantia nigra.^{68,69} NMI has shown greater than 80% sensitivity and specificity to distinguish Parkinson's disease from healthy controls⁷⁰ and could have the potential to show alterations in prodromal Parkinson's disease.⁷¹ QSM assessments of increased iron content in the substantia nigra have shown a broadly similar performance in separating Parkinson's disease from healthy controls.^{72,73} Visual assessment of an area of dorsal nigral hyperintensity, which has been postulated to correspond to nigrosome-1 and is lost in Parkinson's disease, has shown a pooled sensitivity of 98% and a pooled specificity of 95% in distinguishing Parkinson's disease from healthy controls in a meta-analysis of ten case-control studies, including 364 patients with Parkinson's disease and 231 controls,⁷⁴ and has also been suggested as a potential MRI biomarker in prodromal Parkinson's disease.⁷⁵

Although these novel MRI techniques could hold potential as biomarkers of early or even prodromal Parkinson's disease, they generally cannot distinguish between Parkinson's disease and other types of degenerative parkinsonism, since nigral pathology is common to all. This is different for a variety of novel MR diffusion tensor imaging techniques, such as free water imaging and neurite orientation dispersion, and density imaging, which enable differentiation between Parkinson's disease and atypical parkinsonism on the basis of more widespread tissue integrity changes in patients with multiple system atrophy or progressive supranuclear palsy.^{76,77} Some studies have also suggested high discriminative accuracy between Parkinson's disease and multiple system atrophy and progressive supranuclear palsy by use of observer-independent machine learning approaches with automated volumetry or automated voxel-based diffusivity,^{78,79} or multimodal MRI combining several MR parameters.⁸⁰

A variety of radionuclide tracers are available to examine pre-synaptic and post-synaptic striatal dopaminergic function using PET or SPECT imaging.⁸¹ Among these, only dopamine transporter-SPECT has an established role in clinical routine due to its availability and moderate cost. Ligands of the presynaptic monoamine transporter (ioflupane, trodat) used in dopamine transporter-SPECT are sensitive to detect dysfunction or loss of striatal dopaminergic terminals and enable the identification of parkinsonian syndromes with nigral neurodegeneration, including Parkinson's disease and non-degenerative phenocopies, such as essential tremor, psychogenic, or vascular parkinsonism.⁸² Some studies have attempted to use DAT-SPECT to distinguish atypical parkinsonism from Parkinson's disease by measuring the asymmetry

index or the caudate to putamen binding ratio; however, these have not proved to be useful in clinical practice, and DAT-SPECT should not be considered a tool for the differential diagnosis between neurodegenerative parkinsonisms.⁸³ Although dopaminergic radiotracer imaging using DAT-SPECT or metabolic imaging with fluorodeoxyglucose-PET have shown sensitivity for prodromal stages of Parkinson's disease,^{58,84} there is reason to expect that sensitive and specific radiotracer probes enabling visualisation and quantification of α -synuclein deposits in the brain or peripheral autonomic nervous system via PET or SPECT imaging might substantially enhance early or even pre-clinical diagnosis of Parkinson's disease and other synucleinopathies. Several candidates are in preclinical development, but none has yet reached the stage of clinical diagnostic testing.⁸⁵

Fluid and tissue α -synuclein markers

Pathological α -synuclein species are candidates in the search for sensitive and specific biomarkers. A variety of biopsy studies have suggested that immunohistochemical assessment for the presence of phosphorylated and aggregated α -synuclein in the enteric nervous system, and autonomic nerve fibres in the salivary glands or skin can distinguish patients with Parkinson's disease from healthy controls and, more importantly, might serve as a biomarker for prodromal disease stages.^{86–88}

The availability of in vitro conversion assays with ultra-high sensitivity for amyloidogenic proteins like Real-Time Quaking Induced Conversion (RT-QuIC) and Protein Misfolding Cyclic Amplification (PMCA) has substantially effected the search for molecular biomarkers. A number of case-control studies have found sensitivities and specificities of RT-QuIC or PMCA analyses of α -synuclein seeding activity in the CSF of more than 90% to distinguish Parkinson's disease from healthy controls or patients with tauopathies.^{89–92} High sensitivities and specificities have been shown for RT-QuIC or PMCA analyses of skin biopsies.⁹³ Additionally, α -synuclein seeding activity has also been found by use of RT-QuIC in the CSF of non-manifesting carriers of a *LRRK2* mutation⁹⁴ and in the CSF and olfactory mucosa of patients with RBD.^{95,96} In one case-control study that compared 52 patients with RBD with 40 healthy controls, CSF α -synuclein RT-QuIC obtained a sensitivity and specificity of 90%, and the positivity of the test was associated with an increased risk of subsequent diagnosis of a synucleinopathy.⁹⁶ These findings suggest that protein misfolding assays for α -synuclein might play a role in the detection of prodromal or pre-clinical stages of Parkinson's disease. Notably, a recent CSF PMCA study that encompassed 439 samples, including Parkinson's disease (n=71) and multiple system atrophy (n=33) cases, provided evidence for distinctive strains of α -synuclein in Parkinson's disease and multiple system atrophy and thus the potential to distinguish between different synucleinopathies.⁹¹ An overview of the most promising biomarker candidates that could help to

	Diagnostic potential			Disease progression	Comments
	Prodromal Parkinson's disease	Manifest Parkinson's disease	Differential diagnosis (non-Parkinson's disease)		
MRI					
Substantia nigra neuromelanin ^{68,70,71}	Possibly informative	Informative	Not informative	Informative	Insufficiently studied in prodromal Parkinson's disease
Dorsolateral nigral hyperintensity ^{74,75}	Possibly informative	Informative	Not informative	Not informative	Insufficiently studied in prodromal Parkinson's disease; no differentiation between Parkinson's disease and atypical Parkinson's disease
Quantitative Susceptibility Mapping ^{69,72,73}	Unknown	Informative	Not informative	Unknown	Limited number of studies; use in differential diagnosis unclear; no progression data
Tensor imaging (Free Water MRI; NODDI) ^{76,77}	Unknown	Informative	Informative	Unknown	Role in early or prodromal disease unclear; no progression data
Automated volumetry ^{78,79}	Not informative	Not informative	Informative	Informative	Insufficiently studied in prodromal Parkinson's disease
Multimodal MRI ⁸⁰	Unknown	Not informative	Informative	Unknown	Insufficiently studied in prodromal Parkinson's disease; no progression data
Radiotracer imaging					
α-synuclein	Informative	Informative	Insufficient data	Informative	Tracers not yet available
Tau ⁹⁷	Not informative	Not informative	Informative	Not informative	Mostly studied in Alzheimer's disease, progressive supranuclear palsy, and corticobasal degeneration; scarce data for Parkinson's disease
Blood					
Neurofilament ⁹⁸	Not informative	Not informative	Informative	Not informative	Non-specific marker for neurodegeneration; insufficiently studied in prodromal Parkinson's disease
CSF					
α-synuclein seeding activity (RT-QuIC, PMCA) ^{89-92,94,95}	Informative	Informative	Informative	Not informative	Can distinguish different α-synuclein strains in Parkinson's disease vs multiple system atrophy
Neurofilament ⁹⁹	Not informative	Not informative	Informative	Not informative	Non-specific marker for neurodegeneration; can differentiate Parkinson's disease from atypical parkinsonism at group level; insufficiently studied in prodromal Parkinson's disease
Tissue biopsies					
Dermal α-synuclein immunohistochemistry ⁹⁶	Informative	Informative	Not informative	Unknown	Requires multi-site sampling; limited sensitivity, not specific to Parkinson's disease vs other synucleinopathies
Dermal α-synuclein seeding activity (RT-QuIC; PMCA) ⁹³	Unknown	Informative	Informative	Unknown	Limited in vivo information
Olfactory mucosa α-synuclein seeding activity (RT-QuIC) ⁹⁶	Possibly informative	Informative	Unknown	Unknown	Few studies
Gastrointestinal α-synuclein ⁹⁷	Unknown	Insufficient data	Not informative	Unknown	Invasive; requires multi-site sampling; limited sensitivity
Salivary gland α-synuclein ^{87,88}	Informative	Informative	Not informative	Not informative	Invasive; limited sensitivity
Defecation					
Gut microbiota	Unknown	Insufficient data	Not informative	Not informative	Variable results regarding different composition of microbiome between Parkinson's disease and controls
Digital biomarkers					
Multiple motor and non-motor assessments or wearable devices ¹⁰⁰	Possibly informative	Not informative	Not informative	Informative	Insufficiently studied in prodromal Parkinson's disease; no differentiation between Parkinson's disease and atypical Parkinson's disease
NODDI=neurite orientation dispersion and density imaging. RT-QuIC=real-time quaking-induced conversion. PMCA=protein misfolding cyclic amplification.					
Table 3: Candidate biomarkers for Parkinson's disease					

NODDI=neurite orientation dispersion and density imaging, RT-QuIC=real-time quaking-induced conversion, PMCA=protein misfolding cyclic amplification.

Table 3: Candidate biomarkers for Parkinson's disease

identify people at-risk and in the prodromal stages of disease, enhance diagnostic accuracy, and enable monitoring of disease progression is presented in table 3.

Conclusions and future directions

The diagnosis of Parkinson's disease has profound implications for patients and their families, and despite important advances, it remains a challenge. Diagnosis is anchored on well-defined criteria that have shown

excellent sensitivity and specificity in clinical series,⁵⁶ but diagnostic accuracy at a patient's first visit is less than 100%, even when assessed by a movement disorder specialist. This scenario will improve over the next decade as new Parkinson's disease-specific biomarkers become available. Observer-independent machine-learning approaches to MRI data can distinguish Parkinson's disease from its atypical mimics, such as multiple system atrophy or progressive supranuclear palsy,⁷⁸ and further

Search strategy and selection criteria

References for this Review were identified by searches of PubMed between Jan 1, 2006, and Dec 20, 2020, by use of the following terms: parkins*[title] AND "Parkinson's disease", "diagnosis", "epidemiology", "risk factor", "genetic", "premotor", "prodromal", and "atypical". Bibliographies of papers were also reviewed. Papers published in English and German were considered. The final reference list was generated on the basis of relevance to the topics covered in this Review.

advances in imaging markers for Parkinson's disease, including radiotracer imaging of α -synuclein, are on the horizon.⁸⁵

Genetics will play an essential role in the future of Parkinson's disease diagnosis. Additional monogenic forms of Parkinson's disease can still be identified. Aside from monogenic Parkinson's disease, large-scale genome-wide association studies show that the genetic cause in most patients is complex, with multiple susceptibility variants driving disease risk within each patient. Every patient carries numerous genetic risk variants for Parkinson's disease that act synergistically with stochastic and environmental or lifestyle factors to result in development of the disease. These insights give rise to the possibility of using polygenic risk scores or machine-learning algorithms to differentiate patients from healthy controls and to predict patient subgroups, age at onset, and clinical features.^{38,101,102} These advances will also have an effect on the development and application of disease modifying therapies, as already exemplified by ongoing clinical trials targeting the GBA pathway or LRRK2 function.^{15,103}

As soon as disease-modifying interventions become available, there will be pressure to test and apply them to individuals in prodromal stages or to individuals with an increased risk for Parkinson's disease. Such pre-diagnostic stages of Parkinson's disease will only be diagnosed through biomarkers—including imaging, synuclein assays, tissue biopsies, and genetic biomarkers—that are currently being studied.

Parkinson's disease will evolve from a purely clinical to a biomarker-supported diagnostic entity, and new opportunities for early diagnosis will arise, and the diagnostic accuracy at the first neurological consultation will be substantially higher than today. Future generations of neurologists might no longer view Parkinson's disease as a single nosological entity, but will be able to confidently diagnose subtypes with different prognoses and treatment responses.⁴⁰ However, new challenges will arise, particularly when a diagnosis could be made in individuals free of symptoms but without the prospect of preventive therapy. Fortunately, the number of candidate drugs for disease-modification in clinical development has never been as large as today.¹⁰⁴

Contributors

ET and WP led the writing of this manuscript. All authors contributed equally to the literature search, design of tables, panels, and figures, and writing of the Review.

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